

AI BASED RESUME MAKER

Project Report

Abstract

The AI Resume Maker is a web-based application designed to help users quickly create ATS-friendly resumes using structured input and AI-assisted text generation. The system is developed using Flask (Python) for the backend and HTML, CSS, and JavaScript for the frontend.

Users provide details such as personal information, summary, skills, experience, education, projects, and certifications. Based on this input, the system generates a professional resume in plain text format. In cases where AI services are unavailable due to API or network issues, a fallback generator ensures uninterrupted functionality.

The application also supports exporting resumes in multiple formats, including TXT, PDF, DOCX, JPG, and PNG. A modern and responsive user interface enhances usability through smooth transitions, clear status messages, and an intuitive layout.

1. Introduction

In today's competitive job market, a well-structured and ATS-optimized resume is essential. However, many candidates struggle with writing effective content, formatting resumes, and highlighting relevant skills.

The AI Resume Maker addresses these challenges by combining user-provided information with AI-assisted generation to produce structured and professional resumes.

The main goals of this project are:

- **Ease of Use: Simple form-based resume creation**
- **Automation: AI-assisted content generation**
- **Practical Output: Multiple downloadable formats**

This report outlines the system design, implementation, features, testing, and future improvements.

2. Problem Statement

Students and job seekers often face the following problems:

- **Difficulty in writing professional and concise resume content**
- **Lack of role-specific customization**
- **Limited access to free and flexible resume tools**
- **Poor user experience in existing platforms**

Most existing solutions are either paid, complex, or lack reliability. Therefore, a lightweight and efficient resume generator was required.

3. Objectives

The objectives of this project are:

- **Develop a web-based resume generator**
- **Integrate AI-based content generation**
- **Implement fallback generation for reliability**
- **Provide multi-format downloads (PDF, DOCX, JPG, PNG, TXT)**
- **Design a modern and user-friendly interface**
- **Maintain a simple and scalable architecture**

4. Technology Stack

Backend

- **Python 3**
- **Flask**

Frontend

- **HTML5**
- **CSS3**
- **JavaScript**

AI Integration

- **Gemini API (gemini-1.5-flash)**

File Generation Libraries

- **reportlab (PDF generation)**
- **python-docx (DOCX generation)**
- **Pillow (Image generation)**

Development Environment

- **Windows + PowerShell**
- **Localhost server (<http://localhost:5000>)**

5. System Design

5.1 Workflow

- 1. User opens the web application**
- 2. Enters required resume details**
- 3. Clicks “Generate Resume”**

- 4. Data is sent to backend API**
- 5. Resume is generated using:**
 - **AI (Gemini API), or**
 - **Fallback template**
- 6. Output is displayed**
- 7. User downloads or copies the resume**

Page 3

5.2 Core Modules

Input Module

Collects user details:

- **Personal information**
- **Skills and experience**
- **Projects, education, certifications**

Generation Module

- **Creates structured prompts**
- **Uses AI for content generation**
- **Implements fallback logic**

Export Module

Supports:

- **PDF (reportlab)**
- **DOCX (python-docx)**
- **JPG/PNG (Pillow)**
- **TXT (browser download)**

UI Module

- **Modern glassmorphism design**
- **Responsive layout**
- **Smooth animations**
- **Clear status feedback**

6. API Endpoints

GET /

- **Loads main interface**

POST /api/generate-resume

- **Input: User data (JSON)**
- **Output:**
 - **Resume text**
 - **Source (AI or fallback)**

POST /api/download-resume

- **Input:**
 - **Resume text**
 - **Format**
- **Output: Downloadable file**

7. Features Implemented

- **AI-based resume generation**
- **Fallback generation system**
- **Responsive and modern UI**

- **Copy-to-clipboard functionality**
- **Multi-format downloads**
- **Input validation**
- **User-friendly feedback messages**

Page 4

8. Testing and Validation

Testing was performed for:

- **API functionality**
- **Fallback mechanism**
- **File download formats**
- **UI responsiveness**
- **User interactions**

Results:

- **All APIs function correctly**
- **Downloads work across formats**
- **UI performs smoothly across devices**

9. Results

The system successfully:

- **Generates professional resumes quickly**
- **Ensures reliability via fallback**
- **Provides practical export options**
- **Enhances user experience through UI design**

10. Limitations

- **AI output depends on API availability**
- **Resume is plain text (no styled preview)**
- **Image exports have minimal formatting**
- **No user login or cloud storage**

11. Future Scope

Future improvements include:

- **Multiple resume templates**
- **ATS keyword optimization**
- **Advanced PDF styling**
- **User authentication system**
- **Cloud deployment**
- **LinkedIn data integration**

12. Conclusion

The AI Resume Maker successfully achieves its goal of providing a simple, intelligent, and reliable resume generation system. It integrates AI capabilities with fallback logic and multi-format export options, making it suitable for both academic and practical use.

With further enhancements, the system can be developed into a full-scale production-ready application.